

Abdulsalam Oluwaseun Ayeleru

AI Systems Engineer | Full-Stack Software Engineer | Product Engineer

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AI Systems and Full-Stack Software Engineer building production AI, distributed backends, and SaaS products. Owns web, mobile, and cloud delivery, with depth in retrieval, document intelligence, payments, and first-principles machine learning.

Selected Achievements

- Owned **two production environments** for web and Android: GitHub branch strategy, release controls, and environment configuration.
- Published **12+ technical articles** on transformers, NLP, tokenization, inference optimization, and deep learning fundamentals.
- Mentored **200+ students** in AI through Skillsphere; co-led RAIN-INN Robotics & AI Community for two terms.

Core Technical Skills

AI & Machine Learning: LLMs, Retrieval-Augmented Generation (RAG), NLP, transformers, PyTorch, AI agents, computer vision
Backend & Data: Python, TypeScript, JavaScript, SQL, NestJS, FastAPI, REST APIs, PostgreSQL, MongoDB, authentication
Cloud & Systems: AWS Lambda, Amazon SQS, DynamoDB, Docker, GitHub Actions, CI/CD, distributed and event-driven systems
Full-Stack & Product: React, Next.js, Capacitor, Android, Google Play Console, SaaS architecture, product design, release management

Professional Experience

GoodTenants / Upward Ecosystem

March 2026 – Present

Software Engineer & Product Engineer

Hybrid

- Shipped and operated **two production Android applications**, Upward Pay and Upward PM, on Google Play Store; supervised release preparation, testing, Play Console operations, rollout, and maintenance.
- Architected a secure PropTech ecosystem linking tenants, landlords, property managers, payment providers, insurance services, and external property management platforms.
- Engineered **30+ REST APIs** across web, Android, and third-party systems for rent collection, leases, documents, verification, and property operations using NestJS, TypeScript, and PostgreSQL.
- Built payment orchestration for rent, verification, property transactions, and financial workflows: Paystack primary processing with automatic Flutterwave failover for continuity during gateway degradation or outages.
- Owned architecture, UI/UX, frontend, backend, infrastructure, and deployment strategy with Next.js, Capacitor, and GitHub Actions; directed branding, logos, interface systems, SEO, and technical optimization across web and mobile.

Codygo

May 2025 – December 2025

AI Full-Stack Engineer

Remote

- Built enterprise LLM/RAG assistants for business automation with FastAPI, TypeScript, and MongoDB, combining semantic search, contextual memory, and low-latency organizational knowledge access.
- Architected configurable ingestion pipelines for websites, PDFs, knowledge bases, Confluence, Jira, and custom business connectors, including extraction, normalization, hashing, and incremental synchronization for semantic retrieval.
- Orchestrated asynchronous AI processing, background jobs, and distributed execution with AWS Lambda, Amazon SQS, and DynamoDB; containerized services with Docker and contributed to GitHub Actions CI/CD.

Tweakrr

March 2025 – May 2025

AI Engineer & Backend Engineer

Remote

- Architected NLP/RAG document pipelines for parsing, semantic context extraction, unsupported-claim detection, scholarly retrieval, and contextual citation recommendations aligned with academic citation requirements.
- Built Python, FastAPI, and MongoDB services for uploads, authentication, asynchronous processing, document management, formatted in-text citations, and reference-list generation.

Institute of Advanced Medical Research and Training (IAMRAT)

December 2024 – March 2025

Artificial Intelligence Engineering Intern

University College Hospital, Ibadan

- Researched breast cancer subtype classification from histopathological images with ResNet and MobileNet deep convolutional networks; connected predictions to LLM-generated diagnostic explanations for clinician-friendly reporting.
- Built internal FastAPI and React applications for clinical data collection, workflow visualization, and research management.

Selected Engineering Projects

Upward Pay — Production FinTech / PropTech mobile app connecting rent payments, tenant verification, insurance, lease visibility, score boosting, and shareable tenant credit profiles. [Google Play Store](#)

Upward PM — Production property management for landlords and property managers: portfolios, rent tracking, maintenance requests, lease documents, tenant communication, and financial reporting. [Google Play Store](#)

Tweakrr — Public AI citation and document intelligence platform connecting academic writing, document analysis, source discovery, and citation generation to automate research workflows. [Live Platform](#)

Asera — AI learning and research platform combining agentic workflows, memory, RAG, course generation, and personalized study support.

AI Retail Shelf Intelligence — Internal computer vision system for object detection, product recognition, and shelf analysis.

Technical Writing & Open Source

Author of [The Deep End](#): mathematics, deep learning, computer vision, LLMs, NLP, backend architecture, and software engineering.

- Articles: *Tokenization and Encoding with Byte Pair Encoding*; *Implementing Skip-Gram using PyTorch*; *Understanding Transformer Architecture*; *Visualizing Positional Encodings*; *Modern Inference Optimizations: KV Cache and Sliding Window Attention*.
- First-principles ML paper reproductions on GitHub: GPT-style transformers, Byte Pair Encoding, Skip-Gram / Word2Vec, KV cache, self-attention, autoencoders, MediaPipe + Transformer sign language recognition, and YOLO-based retail intelligence.

Research interests: LLMs, information retrieval, deep learning, computer vision, distributed systems, and AI infrastructure.

Leadership, Mentorship & Community

- RAIN-INN Robotics & AI Community Co-Lead, University of Ibadan:** Organized workshops, cohorts, AI/robotics initiatives, and student product showcases.
- Skillsphere AI Mentor:** Guided students on AI skills, projects, learning paths, and career development.
- Departmental Workshop Tutor, Computer Science (2025 session):** Taught practical skills and problem-solving.
- Technical Mentor, TechQing:** Mentored women in technology on technical skills, projects, and careers.
- AI Speaker & Workshop Facilitator:** Spoke at STUMPED AI Student Seminar and the 2026 Become Valuable Conference; ran a practical AI workshop at the latter, encouraging technology careers.

Education & Certifications

University of Ibadan — Bachelor of Science in Computer Science, Nigeria

Expected 2027

Certifications: DeepLearning.AI Deep Learning Specialization; Hugging Face Transformers Course.